

# Sampling, FFT, and the Pre-Lab 1 notebook

Fall 2026 · Prof. Ehsan Roohi

**Lecture and guided Jupyter walkthrough**

# Learning goals

- Explain sample rate, sample interval, record length, and Nyquist frequency
- Run prepared analysis cells in a Jupyter notebook
- Interpret time histories, RMS values, and one-sided FFT spectra
- Recognize aliasing before collecting Lab 1 sound data

# Connection to Lab 1

## PRE-LAB

A known voltage signal contains two pure tones.

Amplitude describes signal size; frequency describes how fast it repeats; phase sets its starting position.

## LABORATORY

A microphone converts changing sound pressure into changing voltage.

Moku:Go stores voltage samples. The waveform shows change in time; the spectrum shows the frequencies present.

# Jupyter notebook structure

## MARKDOWN CELLS

Instructions, equations, and response areas.

Double-click to edit. Use Shift+Enter to render and move forward.

## CODE CELLS

Python commands that define variables, calculate results, and make figures.

Run from top to bottom with Shift+Enter.

# Opening and saving the notebook

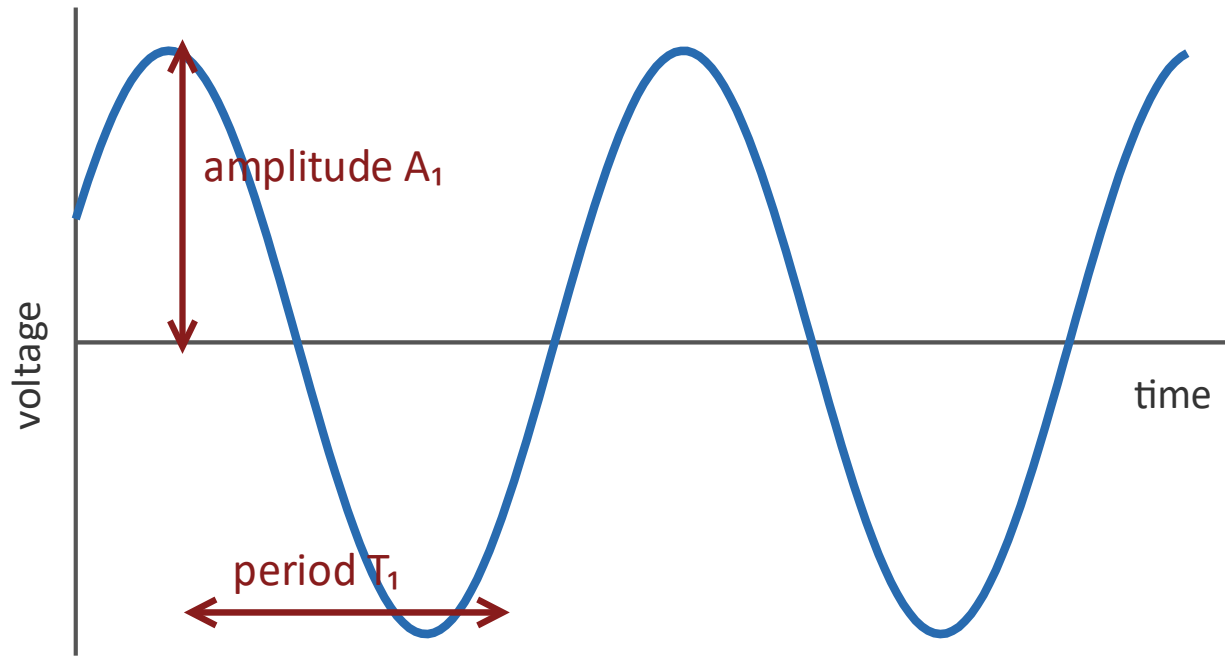
- Download the .ipynb file from Canvas or open it from the course website in Colab
- Save a personal copy before editing
- Run cells from top to bottom
- Keep code outputs and figures in the submitted file

# Execution order matters

```
A = 1.20  
print(A)  
  
# A later cell can use A only after this cell runs.  
rms_component = A / np.sqrt(2)  
print(rms_component)
```

A restart clears stored variables. Run All rebuilds the notebook state from the first cell.

# A sinusoid: what the three parameters mean



$$x_1(t) = A_1 \sin(2\pi f_1 t + \phi_1)$$

$$A_1 = 1.20 \text{ V}$$

Amplitude: peak voltage measured from zero

$$f_1 = 18 \text{ Hz}$$

Frequency: 18 complete cycles each second  
Period  $T_1 = 1/f_1 = 55.6 \text{ ms}$

$$\phi_1 = 25^\circ$$

Phase: where the cycle is at  $t = 0$ ; it shifts the wave left or right

# The new two-tone signal

```
A1, f1, phi1_deg = 1.20, 18.0, 25.0
A2, f2, phi2_deg = 0.45, 42.0, -15.0

def signal(t):
    return (A1*np.sin(2*np.pi*f1*t + np.deg2rad(phi1_deg))
            + A2*np.cos(2*np.pi*f2*t + np.deg2rad(phi2_deg)))
```

Subscript 1 means tone 1; subscript 2 means tone 2. A is in volts, f is in hertz, and phi\_deg is in degrees.

# Meaning of the Python syntax

## ARRAY OPERATIONS

`np.pi` supplies  $\pi$ .

`np.sin` and `np.cos` accept arrays.

`np.deg2rad` converts degrees to radians.

## FUNCTION OUTPUT

`signal(t)` adds the two components at every requested time.

Changing `t` changes the sampling grid, not the physical signal definition.

# Mean and RMS

## MEAN

Average signed level:

$$\text{mean} = (1/N) \sum x[n]$$

Positive and negative parts cancel for complete cycles, so the expected mean is 0 V.

## RMS

Typical signal size, independent of sign:

$$\text{RMS} = \sqrt{(1/N) \sum x[n]^2}$$

For these two distinct tones over complete cycles:

$$\text{RMS} = \sqrt{(A_1^2 + A_2^2)/2} = 0.9062 \text{ V}$$

# Samples and the underlying waveform

## DENSE REFERENCE

The gray curve in the notebook uses 20,001 calculated time points. It shows the mathematical signal clearly.

## 48 HZ SAMPLES

The blue markers contain only 10 measured values in the first 0.20 s. Lines between markers do not add information.

# Sample rate, interval, and record length

```
fs = 126.0                # samples per second
dt = 1/fs                 # seconds between samples
duration = 1.00           # seconds
n = int(round(duration*fs))
t = np.arange(n)/fs

print(dt, n, t[-1])
```

Students read these settings and analyze the resulting data. They do not modify the code.

# Nyquist frequency

- Samples per cycle =  $f_s / f_{\text{signal}}$
- Nyquist frequency:  $f_N = f_s/2$ ; frequencies above  $f_N$  cannot be identified correctly
- For the 42 Hz tone,  $f_s$  must be greater than 84 Hz
- 336 and 126 Hz satisfy the condition. 72 and 48 Hz do not

# Aliasing changes the measured frequency

**FS = 126 HZ**

Nyquist frequency: 63 Hz

Observed FFT peaks: 18 and 42 Hz

Both components remain below Nyquist.

**FS = 48 HZ**

Nyquist frequency: 24 Hz

Observed FFT peaks: 18 and 6 Hz

42 Hz appears at  $|42 - 48| = 6$  Hz.

# The prepared sampling analysis

```
for fs in sample_rates:
    n = int(round(duration*fs))
    t = np.arange(n)/fs
    x = signal(t)
    records[fs] = (t, x)
    sample_mean, sample_rms = mean_and_rms(x)
    print(fs, sample_mean, sample_rms)
```

Students transfer the printed values to an analysis table and calculate RMS percent error.

# Reading the time-domain figures

- Compare sample markers with the dense reference over the same time window
- Look for too few points per cycle and apparent changes in shape
- Read the printed mean and RMS values
- Avoid judging sample quality from smooth connecting lines

# The prepared FFT analysis

```
X = np.fft.rfft(x>window)
magnitude = np.abs(X)/(n*coherent_gain)
magnitude[1:-1] *= 2
frequency = np.fft.rfftfreq(n, d=1/fs)

# Horizontal axis: frequency in Hz
# Peak location: frequency present
# Peak height: estimated voltage amplitude
```

The FFT reorganizes the same samples from voltage versus time into magnitude versus frequency.

# Record duration controls frequency resolution

## SHORT RECORD: $T = 0.25 \text{ s}$

Bin spacing  $\Delta f = 1/T = 4 \text{ Hz}$

The FFT checks frequencies 4 Hz apart. Nearby tones may be hard to separate.

## LONGER RECORD: $T = 1.00 \text{ s}$

Bin spacing  $\Delta f = 1/T = 1 \text{ Hz}$

The FFT checks frequencies 1 Hz apart. Peaks can be located more precisely.

# Expected FFT peaks

## ADEQUATE SAMPLE RATE

336 Hz gives peaks at 18 and 42 Hz.

126 Hz gives peaks at 18 and 42 Hz.

## ALIASING

72 Hz gives peaks at 18 and 30 Hz.

48 Hz gives peaks at 18 and 6 Hz.

# Guided notebook walkthrough

- 1 Define the signal**
- 2 Choose  $f_s$  and create sample times**
- 3 Plot samples beside the reference**
- 4 Compute the one-sided FFT**
- 5 Explain the physical meaning**

## Required data analysis

- Calculate theoretical mean and RMS, then compute RMS percent errors
- Predict every FFT peak before viewing the spectra
- Compare predicted and observed peaks and explain aliasing
- Choose a sample rate and duration for a new measurement and justify both

# Common notebook problems

## CODE PROBLEMS

NameError: run earlier cells.

Missing figure: run the plotting cell.

Changed variable: restart and Run All.

## ANALYSIS PROBLEMS

Wrong units for phase.

Confusing  $f_s$  with signal frequency.

Calling a smooth line measured data.

Reporting peaks without explaining aliasing.

## From the notebook to Moku:Go

- Choose a sample rate from the known or estimated sound frequency
- Record at least one second and save the data as a MAT file
- Plot the measured waveform and compute its spectrum
- Compare the dominant peak with the tuning-fork frequency and replay the recording

# Pre-Lab 1 submission

## BEFORE SUBMITTING

Run All cells.

Keep figures visible.

Complete four response sections.

Export to HTML or PDF if Canvas requests it.

## DEADLINE

Normally posted Monday before the laboratory.

Submit before your own section begins.

Late pre-labs are not accepted.